ClimaSentinel

Live Climate Risk Dashboard & Automated Climate Pipeline

CLIMASENTINEL

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The Mission / Overview

A live climate dashboard for what is happening now.

ClimaSentinel turns daily weather, air-quality, river and climate-baseline signals into a clear **Global Tipping Risk Score** for 10 major European cities. The core product is not a black-box model: it is a live, explainable dashboard that shows the current risk level, the main threat driver and how each city compares at a glance.

10 European cities	**LIVE** risk dashboard	**0** marginal cloud cost

Objectives: Business Impact First

What judges should remember

- **Live situational awareness:** a city can be understood in seconds instead of reading multiple weather, pollution and flood feeds.
- **One score, many signals:** heat, wind, air quality, river discharge and climate baselines become one simple city-level risk story.
- **Decision-ready outputs:** every city receives a risk level, primary threat and plain-language explanation rather than raw technical tables.
- **Automated economics:** scheduled cloud ingestion keeps the dashboard lightweight and repeatable.

User experience

- GREEN safe YELLOW watch RED act
- Premium climate-tech interface with color-coded risks, glowing threat indicators and refined progress bars.

Citizen-friendly · Recruiter-ready · Policy-oriented · Production-minded

Method: Live Climate Intelligence Pipeline

A clean 3-layer climate data factory

1. Collection — BRONZE
Every morning at 06:00 UTC, the system fetches fresh weather, air-quality, flood, ERA5 and CMIP6 data for each city.

2. Refinement — SILVER
Raw feeds are cleaned, deduplicated and standardized in BigQuery so the dashboard uses trusted city signals.

3. Live Insights — GOLD
Business tables produce current score, risk color, dominant threat and rankings for the live dashboard.

Dashboard-first intelligence

- **Current climate status:** the main feature is a live view of today's risk level, main threat and city comparison.
- **Experimental forecast:** the 7-day Random Forest module is a research extension tracked with MLflow, not the main product promise.

Raw APIs → Trusted Data → Live Dashboard

Results: Live Dashboard for Climate Now

What users understand in seconds

Status Is my city safe? Green / Yellow / Red risk level	**Driver** Why is risk rising? Heat, air, wind or river pressure	**Compare** Where is attention needed? 10 cities ranked at a glance

Dashboard value

- **Climate now, not raw data:** judges see the current risk situation instantly across all monitored cities.
- **Readable explanations:** every score is paired with the dominant threat so the result feels useful and trustworthy.
- **Operational confidence:** automated refresh, validation tests and live monitoring protect the public-facing dashboard.

Prediction note: the 7-day model is a **research module**; the core product is the live dashboard for climate now.

LIVE current climate	**10** city views	**06:00** daily refresh

Conclusion: Why ClimaSentinel Matters

From environmental noise to confident action

- ClimaSentinel reframes raw climate feeds as a **live decision product**: understandable, automated and stakeholder-ready.
- The MLOps work makes the dashboard reliable: data ingestion, transformation, versioning, testing, deployment and monitoring all support the live experience.
- The forecast remains an **experimental extension**; the proven value is climate status users can understand today.

"Know the climate risk now; explore what may come next."

Perspectives: Next Evolution

Roadmap for a stronger live product

Scale coverage
Move from 10 cities to more regions, districts and vulnerable local zones.

Send alerts
Email, SMS or municipal webhooks when a risk threshold is crossed.

Explain the score
Plain-language drivers, confidence wording and clear city summaries for non-technical users.

Validate forecasts
Keep the 7-day model experimental until accuracy, drift and safety gates prove it reliable.

Product principle: publish what is reliable now, label predictions honestly, and make every climate signal easy to act on.

KEY REFERENCES

[1] Open-Meteo: weather, air quality, flood, ERA5 and CMIP6 APIs.

[2] Google Cloud, BigQuery, dbt, FastAPI, Next.js and monitoring stack.

[3] ClimaSentinel GitHub repository, source code and project documentation.

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Built as a production-style climate intelligence platform for the MLOps final project.

MORE INFORMATION

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